My School and the School Network

Year 8 Information and Communications Technology



Welcome to KS3 ICT

<u>Year 8</u>
Introductions to ICT + Activities
<u>Year 9</u>
Some new packages/applications + Website Design
<u>Year 10</u>
Under the Hood of a Computer + Introduction to Programming

"You don't have to be great to start, but you have to start to be

"You can't have a better tomorrow if you're still thinking about yesterday."

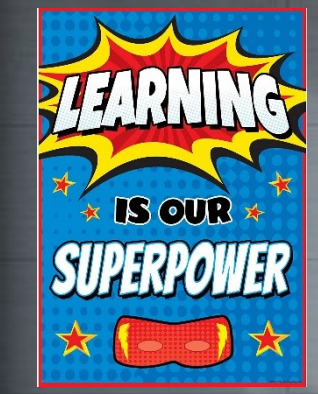
Charles F Kettering

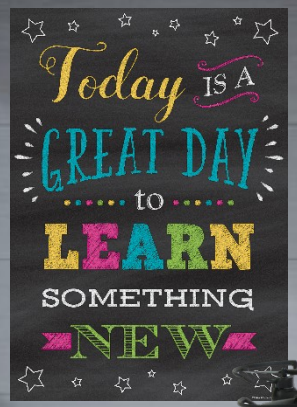


Why Study ICT?

Within KS3 ICT, you're going to learn a whole lot about the world of computers – far more than just how to turn on a games console or use an app on a phone.

ICT in Secondary School is going to teach you about what actually goes on behind the scenes of a computer – and also introduce you to programming your own computer!



















Year 8 ICT

INTRODUCTION

- My School
- Google Classroom
- ICT Contracts
- File Management

UNDER THE HOOD

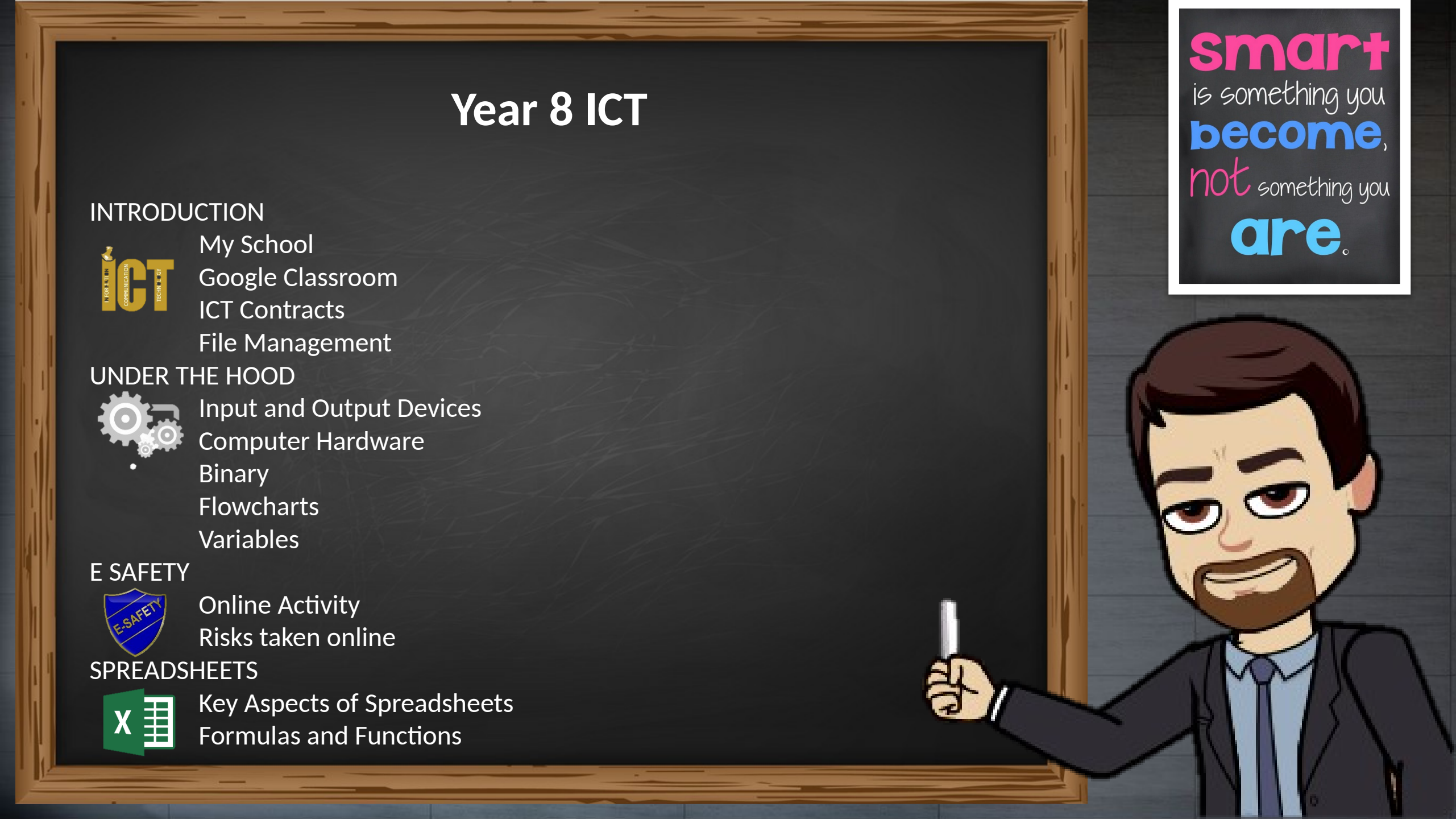
- Input and Output Devices
- Computer Hardware
- Binary
- Flowcharts
- Variables

E SAFETY

- Online Activity
- Risks taken online

SPREADSHEETS

- Key Aspects of Spreadsheets
- Formulas and Functions



Google Classroom

I will be uploading lessons, handouts and homework through Google Classroom. It is important you take note of my email in case you are unable to access resources.



MR BUTLER
ICT

Acceptable Use of Our Computers

- It is very important that we all use the ***School Network*** for our education.
- All users of the network are expected to act responsibly and to show consideration to others.
- Using the school network for non educational purposes will bring about sanctions.

Acceptable Use Policy

- Never tell anyone your password.
- Users should not logon to or use any account other than their own and should logoff when leaving a workstation.
- It is not acceptable to: to download, store or install software to school computers.
- It is not acceptable to:
  - Attempt to introduce a virus or malicious code to the network.
  - Access, download, create, store or transmit material that; is indecent or obscene and brings the name of the school into disrepute.
- The use of public chat facilities and social networks is not permitted.
- Users should not directly copy and use material from the Internet in their school work.

Top 10 Rules of the Computer Room



The School Computer Contract

- Before you can use the computers within school, you need to first agree to some terms and conditions of use.

- What rules do you think we should include that we have just discussed? Are they all within this contract?



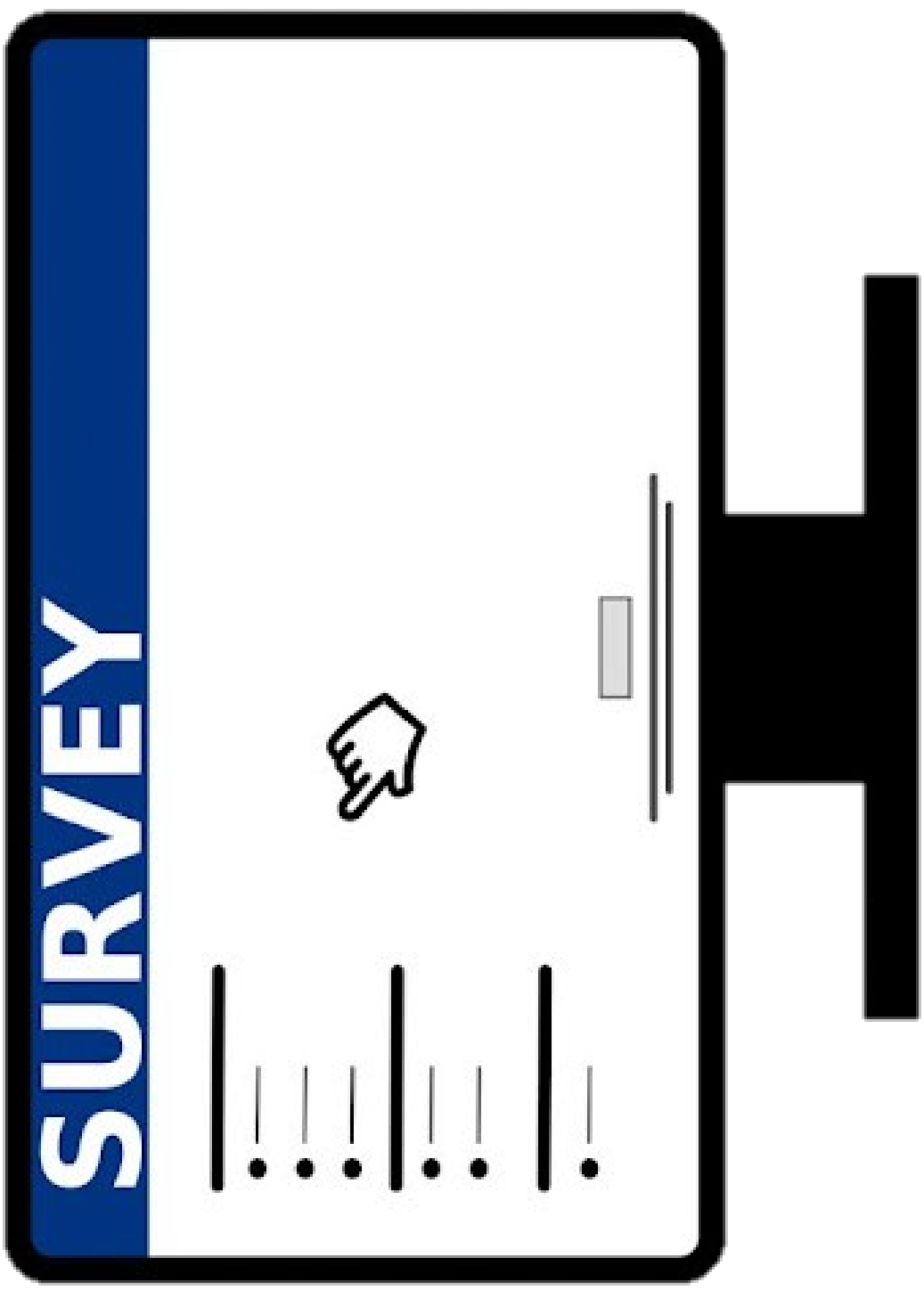
Note: Absence from Class

- If you are absent for any reason from an ICT lesson, it is **your responsibility** to catch up before next lesson.

- This includes:
  - Checking if any homework is due (and needs to be completed)
  - Catch up on any work/notes missed. If you do not understand what you have missed, see me before the next lesson.
  - If there was a class test, you need to see me **before** next lesson to complete this or so I can arrange a time for you to do the test.

- **I will not accept 'I wasn't here last lesson' as an excuse for not completing homework.**

Pupil Baseline Survey



Q1. What is a computer?



- A Computer is a type of machine.
- It is used for storing and processing data according to instructions given to it in a variable program.
- It doesn’t have a brain like humans so can’t think or have ideas, but can do lots of useful things.

What is a Network?



- A **network** is made when two or more computers are joined together.
- This means they can communicate with each other, and also share things like printers and files.
- A computer which doesn’t share or broadcast data is not a network.

What Computer Networks have you used before?

- You have used different **types** of networks. You may have already used:

Social Networks

- So how many here use:
  - Facebook?
  - Twitter?
  - Instragram?
  - Snapchat?
  - Tik-Tok?
  - Pinterest?



The Internet

Home Wifi



Gaming Networks

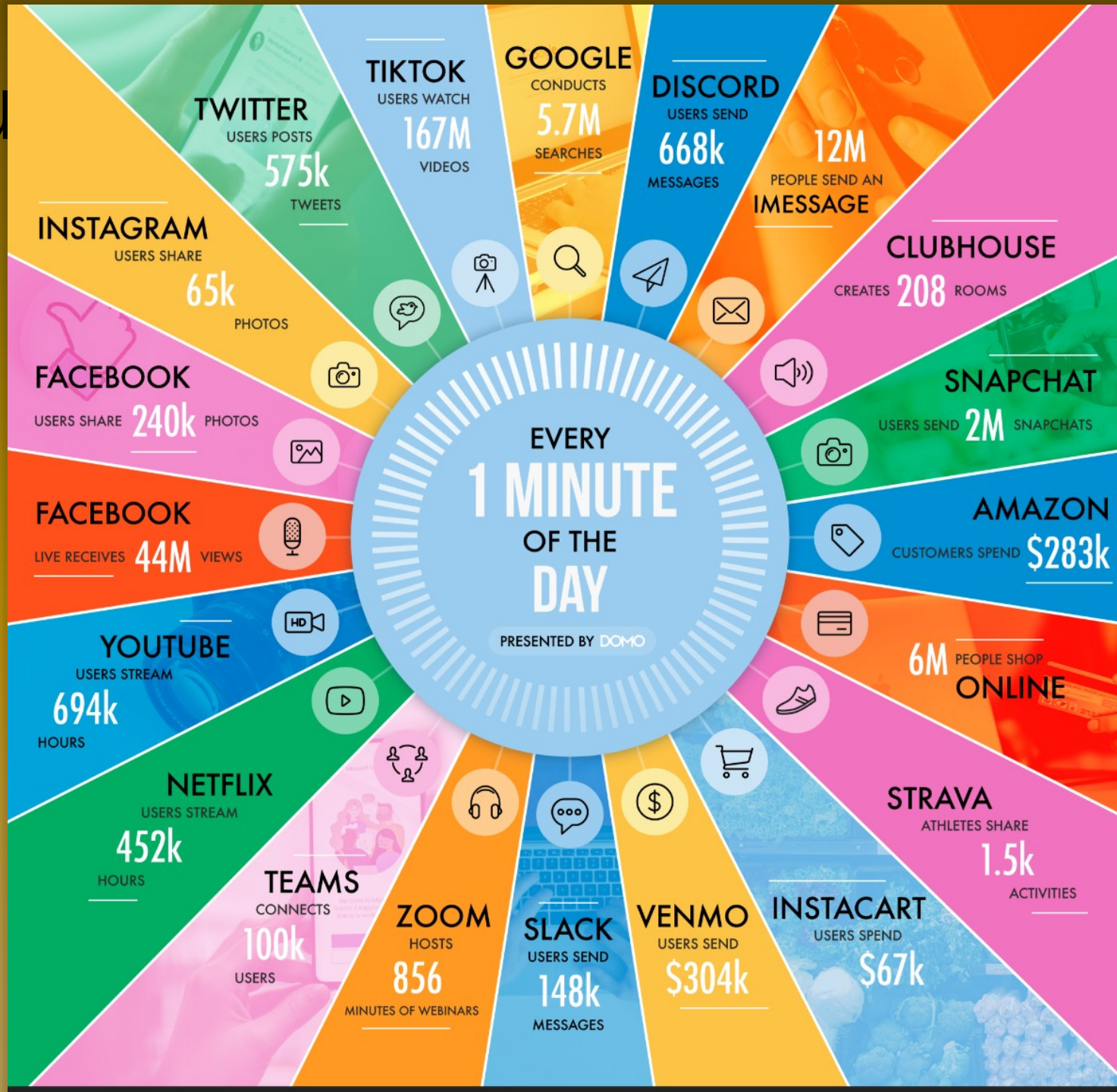
The Internet

- The ***internet*** is a world wide network. It allows computers from all over the world to communicate with each other, such as sending email, online gaming and social networks.



Do you have a minu

- This is what happens during an internet minute in 2021.

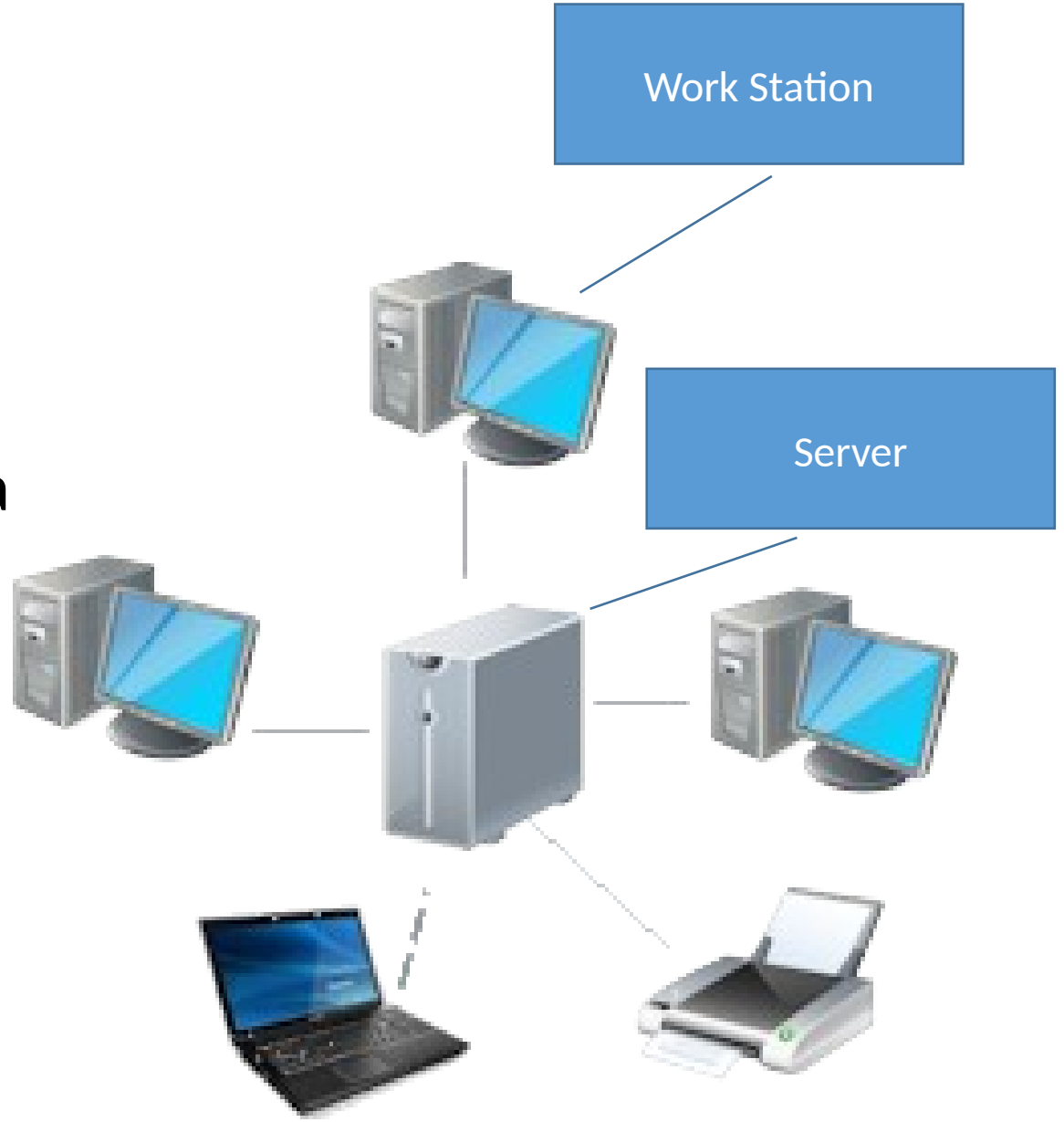


So who owns the internet?

- Nobody is in charge of the internet, which means that anybody can put whatever they want onto it.
- This can cause problems as offensive material can appear on the internet.
- Fortunately, the police can track and identify everything that is posted as the Internet keeps track of all the computers connected to it.

The School Network

- All the computers on our school network are controlled by a main computer called a “***Server***”.
- This allows people to save their files centrally and share things like printers.
- The computer you are using is called a “***Work Station***”.



Accessing a network

- All of us can use the school network – it's called the c2k network that runs throughout Northern Ireland.
- We each have login details to tell the server who we are. These login details include:
  - A **User ID** – this is unique and no one else will have the same ID.
  - A **Password** – this is chosen by you and only you should know it**.** This makes sure you are who you say you are.

Why is it important to keep this information secure and not to share it?

Setting a Strong Password

- We must ensure we have a **strong** password.
- A strong password will include:
  - At least 8 characters – Some capitol and some lower case letters
  - Use some numbers
  - Use symbols like $, %, !, etc
- It should **not** be:
  - Easy to guess
  - Be just a name
  - The same as your username
  - The same as any other password you have used.
- Throughout the year, you may be asked to change your password.

Questions



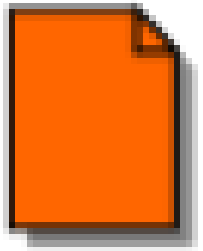
1. What is a **network?** [1]
2. What is a **server?** [1]
3. What is the **Internet?** [1]
4. Why does everyone need a unique **Log In** for the school network? [1]
5. What should be in a **strong password?** [5]
6. What activities are not acceptable on the school network? [7]
7. What could happen if I break the rules? [1]

File Management

Files and Folders

Why is it important to organise files?

1. Much easier to locate files
   - always know where to find what you need
   - You will never waste time wondering where to save anything
   - Can easily direct someone/teacher to access a file for you
2. Easier to keep backups
3. No files need to be saved in several places
   - No need to save files multiple times as you can’t find the first one you saved!
4. More efficient
5. If time is a limiting factor, can find file quickly



Marks.xlsx

Using Em  arch Engines

U01L03 | Mr Butler

Just Some of the Rules!

- No Eating or Drinking in a Computer Suite (including chewing gum!).
- Do not change any settings on the computer
- You must not send inappropriate or upsetting messages – this is cyber bullying.
- Report any damage to the teacher
- Do not touch somebody else's computer.
- Do not play games or surf the internet whilst a lesson is taking place
- Only use Google Classroom for Educational purposes

So – What is an email?

- **Email** stands for ‘electronic mail’
- It is a message sent from one computer to another over the internet.
- In order to use email, you need:
  - An email account (Gmail/Yahoo/AOL/Hotmail etc)
  - Email software (Outlook/Webmail/Thunderbird etc)
  - Internet Access (BT/Virgin/TalkTalk/Plusnet etc)

Are there rules to using Email?

- Just like sending a letter in real life, sending an email follows the same rules.
- It is a **formal** way to message someone, so should be professional.
- **Netiquette** is a term used to show how you should behave when sending emails and using the internet.
  - Be polite
  - Never forget the subject line – who ever receives your email needs to know what it is about
  - Do not type all in capitals as this looks like shouting
  - Don't use "text language"
  - Let people know if you are joking by using a smiley face :) Emoticons!

The People involved in emails

- There are two main groups when sending emails.

SENDER – the person who sends the email

RECIPIENT – the person who receives the email

- When a message is sent, it appears in the **recipient's inbox**
- The recipient will need to open the email to read it



1. Sender Sends Email

4. Recipient can read the message.





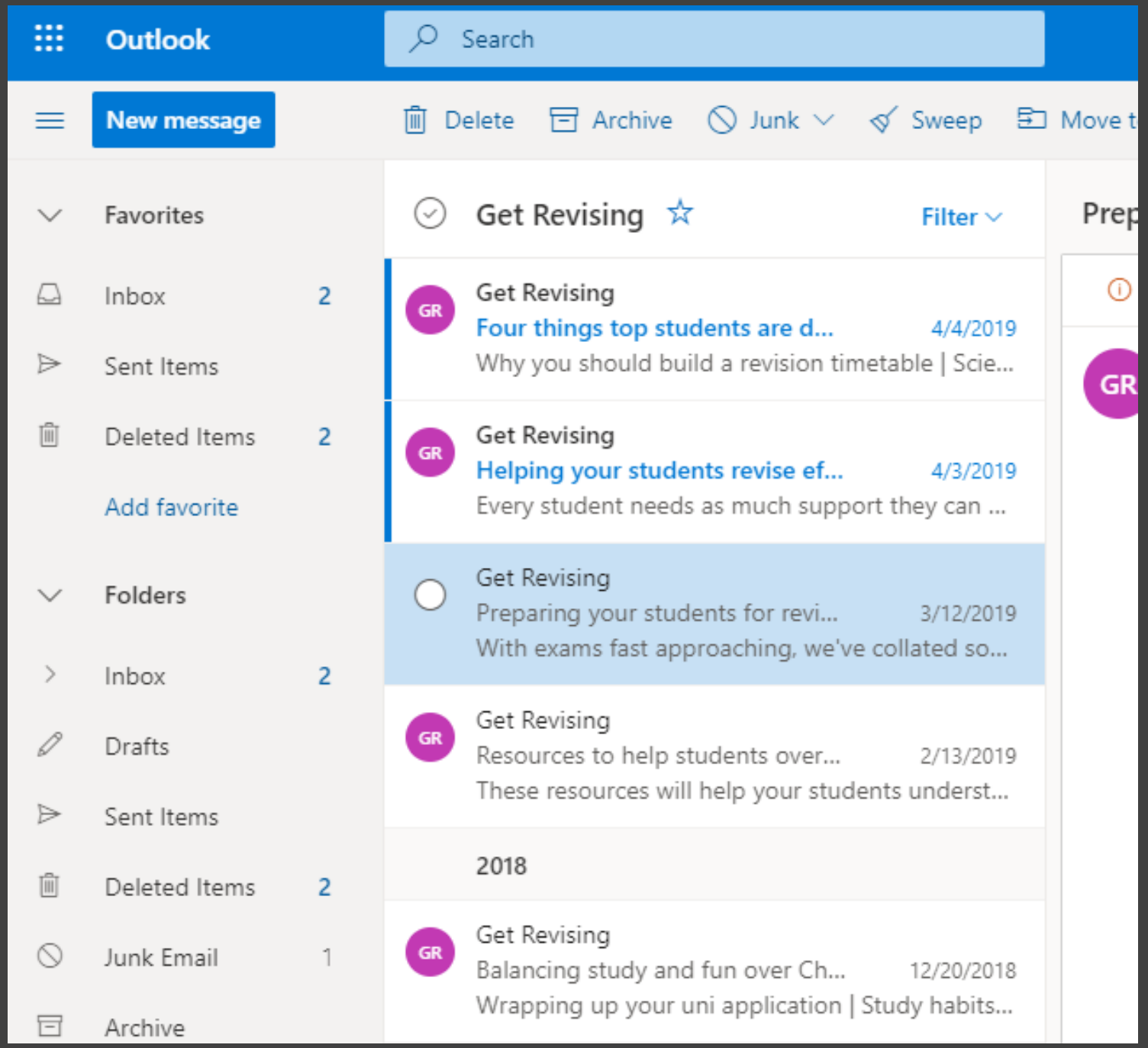
3. Recipient Can open the email from their inbox



2. Email goes into Recipient's inbox

School Email

- We use Microsoft Outlook.
- Things to note for new emails:
  - **New** (unread) emails come into our inbox and have a blue bar
  - The **Sender** appears in **bold**
  - The **Subject** appears in blue
  - A **preview** of the message appears under the subject



Before we send an email:

- We need to know the persons **email address** (where we are sending to)
  - **YOUR** email address uses your username that you used to log in to the school computer.

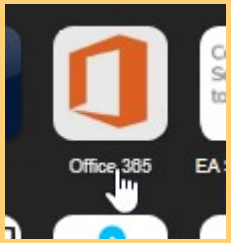
    aperson123
  - Followed by **@c2ken.net**
  - This makes the full email address of **aperson123@c2ken.net**

TIP: Take a note of your email address in your planner!

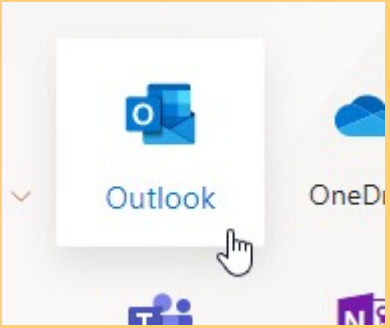
Sending an email



- Open The Internet and go to MySchool
- Click on **Office365**



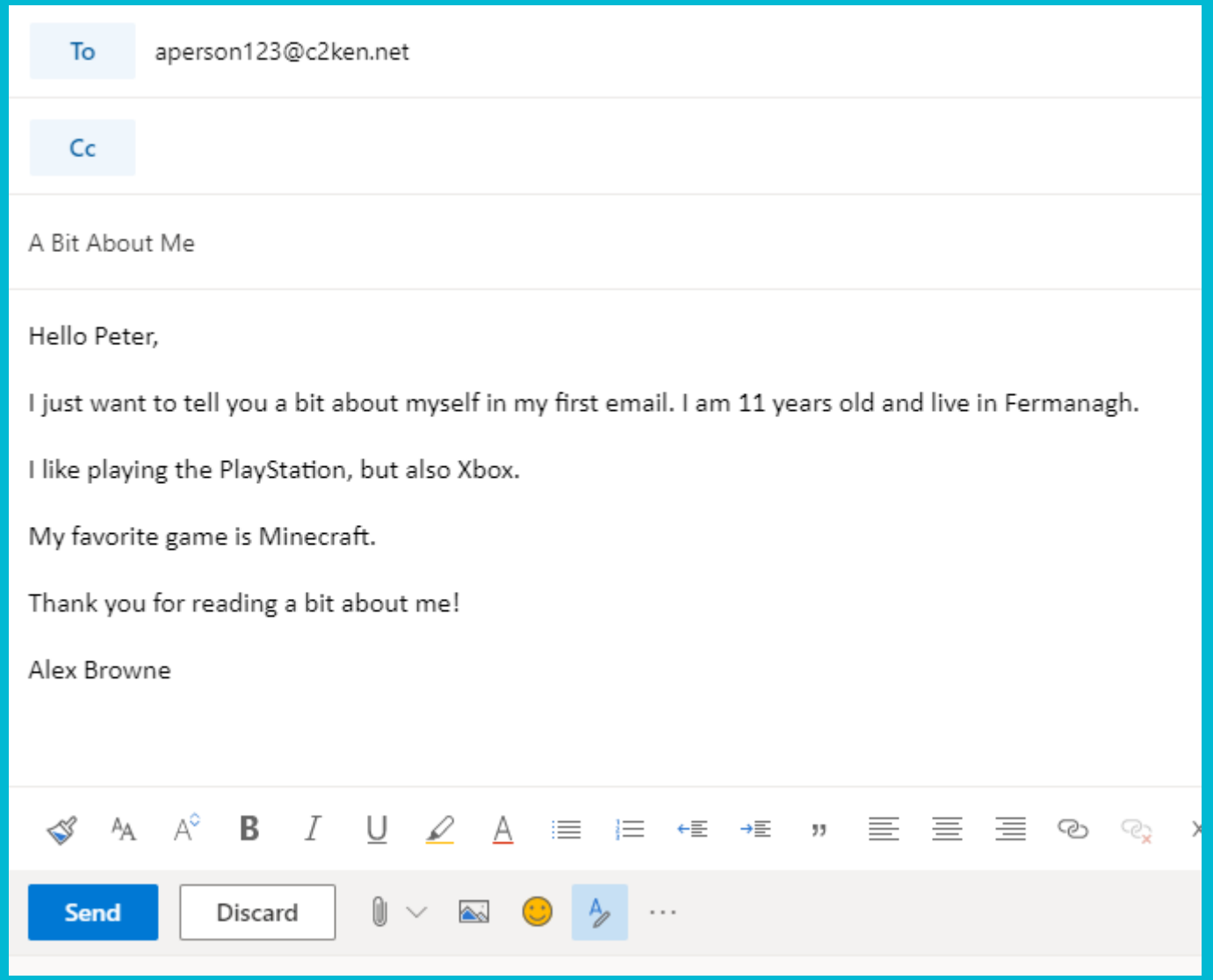
- Select **Outlook**



- Select **New Message**



- Type in the email address of a friend in this class
- Add a meaningful subject line.
- Add a Message using the correct language
- Type in a suitable message to tell the recipient a little about yourself.
  - Your favourite team
  - A film you have recently seen or music artist you like
  - A game you recently played.
- Make sure you sign the email off to say who it's from!
- Click on **Send.**



Receiving email

- When you receive an email, it will appear in your inbox.
- Open your inbox and view your new message from a friend!
- You will be able to see when it was sent, the subject (what it is about) and also be able to preview the start of the message!
- Double click on the email to open and read it in full.

Replying to an Email

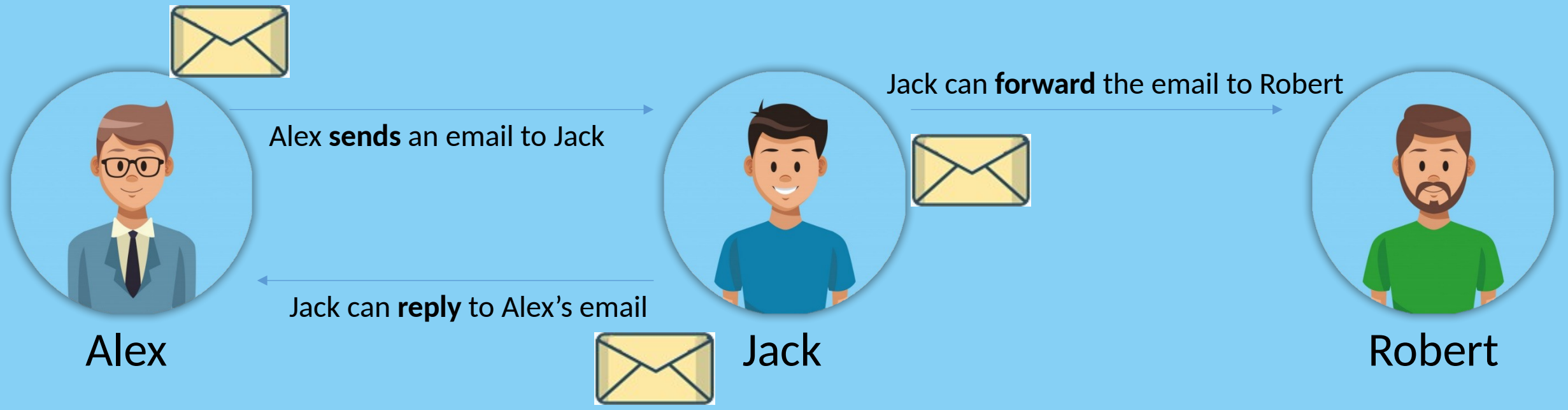
- Once you receive an email, you will also be able to **reply** to the message.
- When you ***reply*** to an email it means you are sending an email back to the person who originally sent it to you.
- Because the message is going back to the person who sent it to you, the email address is automatically entered
- **RE** appears in front of the subject to tell you it's a reply



Forwarding Emails



- We can also forward an email.
- This is where you receive an email from one person, and instead of sending a message back, you send the message onto someone else
- **FW** in the subject tells you it has been **forwarded**


Alex **sends** an email to Jack
Jack can **forward** the email to Robert
Jack can **reply** to Alex's email
Alex
Jack
Robert



Internet and the WWW

The Internet



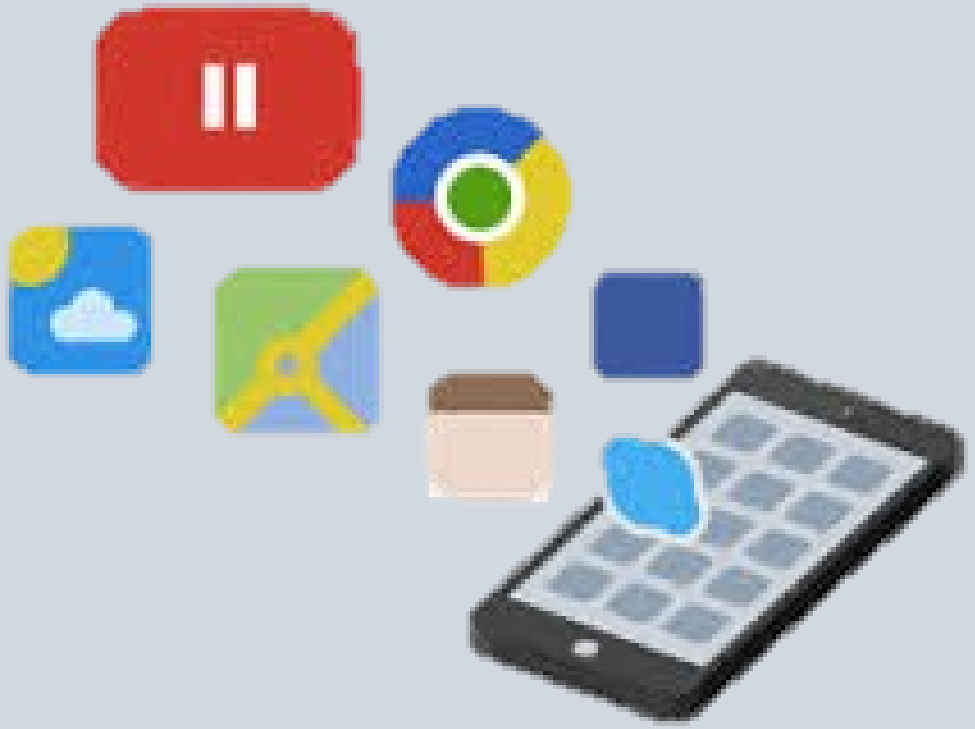
- The ***Internet*** is a massive global network of computers linked together.
- It is physical **hardware** that is connected together.

The World Wide Web

- The ***World Wide Web*** or ***WWW*** is the collection of millions of websites that sits on within the Internet.

- We access the ***WWW*** through a web browser like Google Chrome, Firefox or Internet Explorer.

- It is the **software** on the internet.



How does the Internet work?



A computer asks for information (a website).



The information is sent in something called a packet (like an enevelope with a letter in).



This is sent through miles of underground copper cables.



Through small Regional Networks.



To England's main Internet Hub, Telehouse North, in London.



They pass the information (the packet) as light through miles of under-sea fibre optic cables.



To other countries' main Internet Hubs.



Who pass it to their Regional Networks.



Along their underground copper cables.



To the website's server (a computer which has the website on it).



The server breaks the website down into hundreds and thousands of smaller packets and sends them off.



Back along their underground copper cables.



Back through their Regional Networks.



Back to their main Internet Hub.





Where it's turned back into light and back through the miles of under-sea cables.



Back to London's Internet Hub, Telehouse North.



Back through our Regional Networks.



Back through our underground copper cables.



Where all of the seperate packets meet back at your computer and load up the website.

Search Engines

Search Engines like Google use complex Algorithms to do three tasks:

- **Crawling** – where websites are discovered;
- **Indexing** - where a website is analysed and stored in huge database;
- **Retrieval** – You receive a list of relevant web pages on the topic you are searching for.



Searching the WWW

- 6.51 billion pages on the WWW.
- The size of the World Wide Web (The Internet) The Indexed Web contains at least 6.51 billion **pages** (Thursday, 13 February, 2020).

- **Google** says that it stores information about those 30 trillion **pages** in the **Google Index**, which is now at 100 million gigabytes (2013). It has grown almost exponentially since.

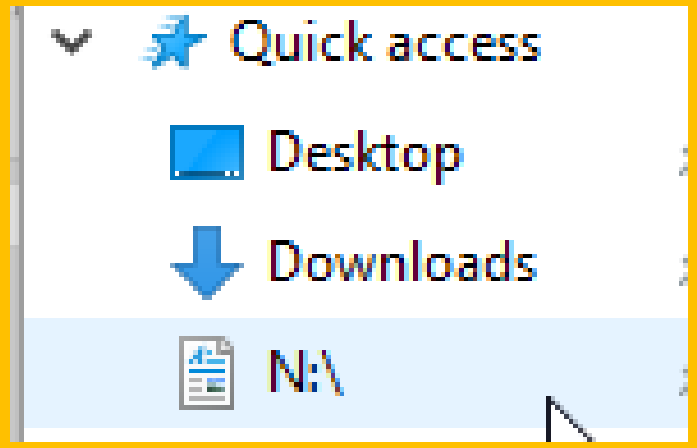


Web Searches

- When we use a **search engine** such as Google or Bing, their web crawler creates a list of search results. These contain **a list of relevant webpages**, with the most relevant at the top.

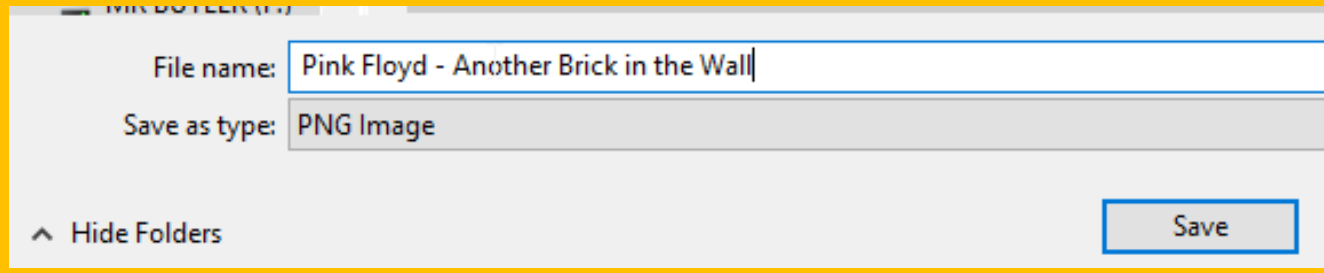


- Click on the **N:/** (the N Drive) to get to your area where you can save work.



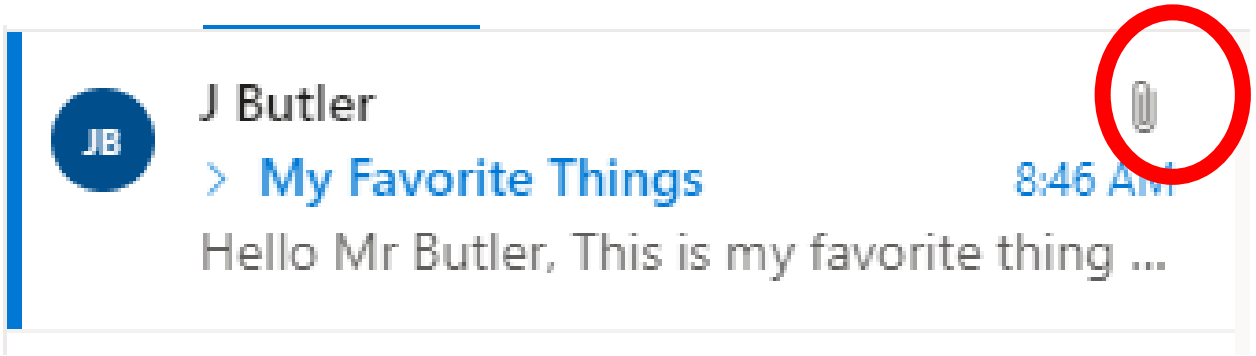
- Then Go to Year 8 > ICT > Unit 1

- If you don't have this file directory, you need to create one!



- Give it a meaningful name and then click SAVE

Receiving an Attachment



- We can also receive an email with an attachment from someone.
- When we receive an email which has an attachment, the paperclip icon appears next to the message.
- You can then open the email and also open or save the attachment!

What is a Computer Virus?

- A virus is computer program designed to deliberately damage or make a computer system unreliable by slowing down the processor.

- They have been created to cause damage to your computer, including:
  - Corrupting files and folders
  - Deleting Data
  - Stealing your personal information
  - Infecting your computer so it slows down/stops working



Avoiding Virus'

- Do not open email attachments from people you do not know.
- Install anti virus software on your computer and make sure you keep it up-to-date.
- Do not download files from the internet unless it is from a known safe site and you have permission from a parent or guardian.
- Scan USB memory sticks for viruses.

